

PROJECT - 1

DEEP LEARNING BASED DRUG–DRUG INTERACTION PREDICTION

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INTRODUCTION

PHARMACOLOGICAL FOUNDATIONS

- **Pharmacovigilance** Monitoring and preventing Adverse Drug Events (ADEs)
- **Polypharmacy** Use of multiple medications by a patient
- **Drug–Drug Interactions (DDIs)** One drug alters the effect, metabolism, or toxicity of another

CLINICAL CHALLENGES

- More medications → Higher risk of DDIs and adverse effects

WHY IT MATTERS

- 20%+ of elderly patients take 10+ medications
- 195K hospitalizations and 74K ER visits/year linked to DDIs
- \$177B annual healthcare burden in the US

PROBLEM STATEMENT

Key Research Gaps

1. Limited Higher-Order Modeling

Most existing models mainly capture interactions between two drugs and struggle to represent the combined effect of 3 or more drugs. Drugs may be safe in pairs but **harmful when combined together**. Pairwise models can miss this effect.

2. Simple Feature Fusion

Drug features are often combined with equal importance, ignoring that each drug may contribute through different biological mechanisms.

3. Severe Class Imbalance

Side-effect datasets have many common effects but very few examples of rare and severe adverse events, making them difficult to predict.

PROBLEM STATEMENT

Existing DDI prediction models mainly focus on pairwise drug relationships, limiting their ability to capture complex, non-additive interactions among multiple drugs, especially rare and severe adverse effects.

LITERATURE REVIEW

- **Deep Learning:** DNN-based models use drug structure and multiple drug features for DDI prediction.
- **Attention Mechanisms:** Improve feature fusion by identifying important drug features.
- **Graph Neural Networks:** Capture molecular structure, substructures, and drug–drug relationships.
- **Knowledge Graphs:** Integrate biomedical information such as drugs, targets, genes, and diseases.
- **Multi-Feature Fusion:** Combining diverse drug information generally improves representation quality.
- **Key Challenges:** Data imbalance, sparse interactions, interpretability, and generalization to new drugs.
- **Research Gap:** Most existing methods focus on pairwise (two-drug) interactions.
- **Motivation:** HyperAttDDI extends these ideas to higher-order DDI prediction involving multiple drugs using hypergraph attention.

LITERATURE REVIEW

Method	Approach	Advantage	Limitation
DeepDDI [1]	Uses DNN with drug names and molecular structures to predict DDIs and drug-food interactions	Works with basic drug information and predicts multiple interaction types	Performance could improve with larger DDI training datasets
DDIMDL[2]	Combines four drug-feature sub-models through a joint DNN for cross-modality learning	Uses diverse drug features for improved DDI event prediction	Class imbalance and rare interactions may cause underfitting
DANN-DDI[3]	Uses attention neural networks to learn drug-pair representations and feature importance	Combines multiple drug information sources and predicts novel DDIs	Noise, class imbalance, and limited interpretability remain challenges
BioChemDDI[4]	Integrates biochemical and network features using self-attention and graph information	Captures both biochemical properties and network structure	Better negative-sample selection is needed to reduce noise
SSI-DDI[5]	Learns pairwise interactions between drug substructures from molecular graphs	Directly learns substructures and provides more interpretable predictions	Sensitive to drug ordering and noise during substructure extraction

LITERATURE REVIEW

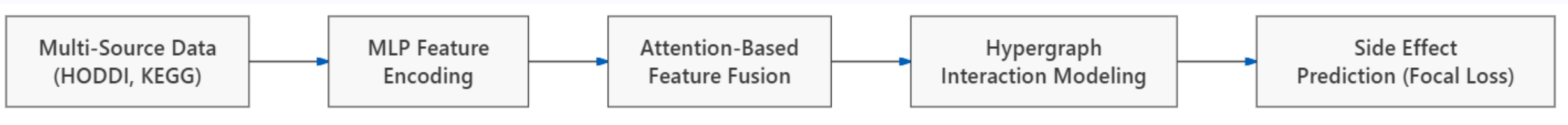
Method	Approach	Advantage	Limitation
DSN-DDI[6]	Learns local/global representations and intra-view/inter-view drug substructures	Provides a generalized framework for real-world DDI applications	Generalization to unseen drugs remains challenging
MIRACLE[7]	Uses unsupervised contrastive learning to integrate multiple molecular views	Captures relationships within and between molecular views effectively	Performance decreases as dataset size increases
MFFGNN[8]	Extracts global molecular-graph and local atomic-level features	Fuses molecular topology, drug interactions, and chemical context	Cannot directly handle multi-type DDI prediction
DDI-MDAE[9]	Learns unified drug representations from multiple drug-feature networks	Handles large-scale, noisy, sparse, and incomplete information	Network topology alone may not provide rich drug representations
AMDE[10]	Encodes drug information from SMILES sequences and atomic molecular graphs	Combines sequence-level and graph-level drug features	Additional drug features could further improve prediction

LITERATURE REVIEW

Method	Approach	Advantage	Limitation
NMDADNN[11]	Integrates five heterogeneous drug-related information sources into unified features	Effectively combines multiple sources of drug information	Similarity matrices could benefit from additional sources and improved measures
DPDDI[12]	Uses GCN to extract DDI network features followed by DNN prediction	Predicts DDIs using network information without requiring drug properties	Cannot predict interactions involving completely new drugs
MDF-SA-DDI[13]	Combines two drugs in four ways to model unobserved interactions	Multi-source drug fusion improves DDI prediction	New-drug prediction performance still needs improvement
KGNN[14]	Uses knowledge graphs to explore drug topological neighborhoods	Captures high-order structural and semantic relationships	Additional drug features may improve prediction accuracy
RANEDDI[15]	Uses relation-aware network embedding for multi-relational drug graphs	Learns richer embeddings by considering different relationship types	Performs poorly with sparse networks and low-frequency DDI types

DESIGN OF PROPOSED METHODOLOGY

- We propose a deep learning pipeline that predicts adverse side effects caused by combinations of **3 or more drugs** taken together.
- Each drug is represented using **four different feature types** extracted from public drug databases.
- These features are independently encoded, then fused using learned attention weights before being passed into a hypergraph-based model.
- The model is trained with Focal Loss to handle severe class imbalance across 4,569 side effect types.



DESIGN OF PROPOSED METHODOLOGY

Multi-Feature Drug Representation

- Each drug is converted into a 2,043-dimensional numerical vector by combining four distinct feature types.
- **Structure features (881 dim):** Binary fingerprint capturing the drug's molecular shape (generated from its SMILES string using RDKit).
- **Target features (780 dim):** Binary vector indicating which proteins in the body the drug interacts with.
- **Enzyme features (129 dim):** Binary vector indicating which liver enzymes are responsible for breaking the drug down.
- **Pathway features (253 dim):** Binary vector indicating which biological processes the drug participates in.

DESIGN OF PROPOSED METHODOLOGY

Attention-Based Feature Fusion

- Existing methods concatenate all four feature types with equal weight, ignoring that different drugs interact through different mechanisms.
- Our approach uses a learned attention mechanism that assigns per-drug importance weights across the four feature types.
- The output is a single compact embedding per drug, where the most relevant feature type is automatically amplified.
- This is a **per-drug** mechanism: it captures each drug's most important characteristics before any combination level modeling.

DESIGN OF PROPOSED METHODOLOGY

Hypergraph-Based Interaction Modeling

- Standard graph neural networks can only model pairwise (2-drug) interactions, they cannot capture joint effects of 3+ drugs.
- We use a hypergraph where each hyperedge groups all drugs in a patient's prescription, natively modeling higher-order combinations.
- An attention mechanism within the hypergraph learns which drugs in a combination are the primary contributors to the predicted side effect.
- Two layers of message passing allow drugs to exchange information through shared combinations.

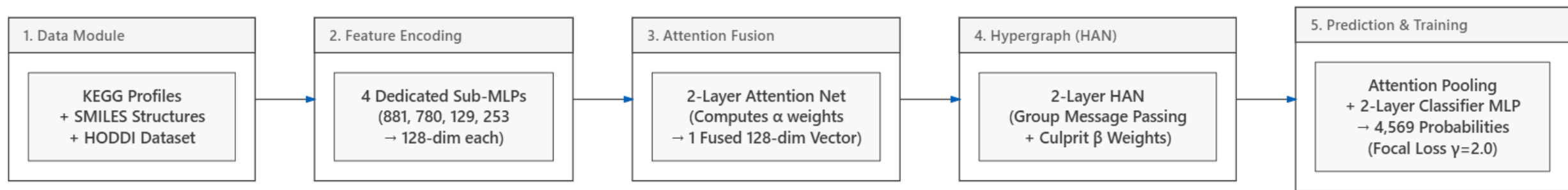
DESIGN OF PROPOSED METHODOLOGY

Prediction and Evaluation Strategy

- The drug combination is represented as a fixed-size vector and classified against 4,569 possible side effects using a multi-label prediction head.
- Training uses **Focal Loss** to down-weight common, easy predictions and force learning on rare but severe side effects.
- Evaluation uses 5-fold stratified cross-validation with Accuracy, Macro-F1, Micro-F1, ROC-AUC, and PR-AUC.
- Four ablation studies isolate the contribution of each novel component: attention fusion, hypergraph structure, focal loss, and multi-feature input.

System Design

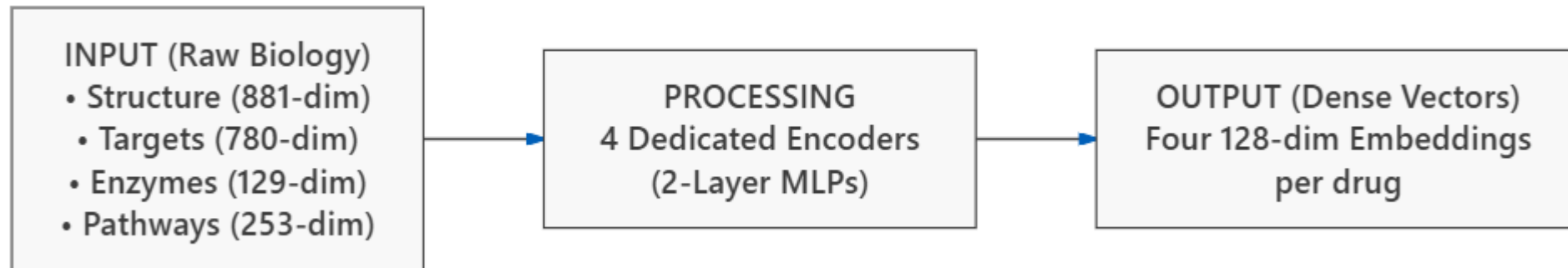
- **Pipeline Architecture:** Built as an end-to-end framework organized into **5 technical modules** (Data Preprocessing, Feature Encoding, Attention Fusion, Hypergraph Interaction, and Prediction & Training).
- **Data Ingestion (M1):** Ingests individual drug features from **KEGG & SMILES** and real-world multi-drug prescriptions from **HODDI**.
- **Neural Components (M2–M4):** Integrates **4 Encoder MLPs (M2)**, a **2-layer Attention Fusion Network (M3)**, and a **2-layer Hypergraph Attention Network (M4)**.
- **Prediction & Output (M5):** A **Classification MLP** outputs probabilities across **4,569 side effect classes** with **attention weights** for culprit drug identification.



Module Description

Data Preprocessing & Feature Encoding Module

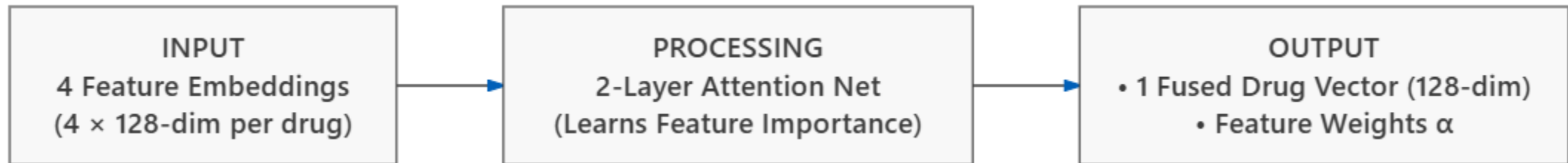
- Purpose: Converts raw drug data into standardized 128-dimensional dense vectors.
- Input: 2,043 raw binary features per drug (molecular structure, protein targets, liver enzymes, and biological pathways).
- Processing: Four separate 2-layer Multi-Layer Perceptrons (MLPs) independently encode each feature type using Linear, ReLU, and Dropout layers.
- Output: Four separate 128-dimensional embedding vectors per drug, passed to the Fusion Module.



Module Description

Attention Based Fusion Module

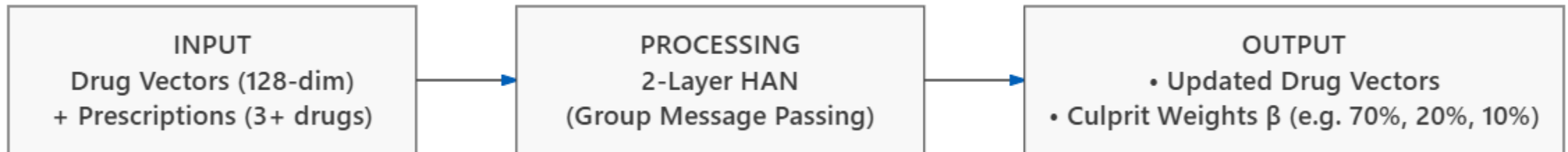
- **Purpose:** Combines the four feature vectors into a single, comprehensive embedding per drug.
- **Input:** The four 128-dimensional feature vectors produced by the encoder MLPs.
- **Processing:** A 2-layer Attention Network calculates an importance weight for each feature and performs a weighted sum.
- **Output:** One fused 128-dimensional embedding per drug and feature-level attention weights showing which mechanism matters most.



Module Description

Hypergraph Interaction Module:

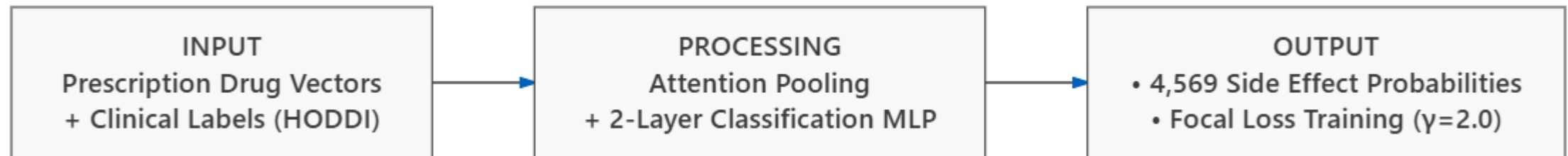
- Purpose: Models complex group interactions among 3 or more co-administered drugs.
- Input: Fused drug embeddings and an incidence matrix defining which drugs appear in each prescription.
- Processing: A 2-layer Hypergraph Attention Network (HAN) performs message passing and computes drug-level attention weights to identify the main culprit drug.
- Output: Context-aware drug embeddings and clinical culprit scores (e.g., weights [0.70, 0.20, 0.10] for a 3-drug combo).



Module Description

Prediction Module & Training Pipeline

- Purpose: Aggregates group features, predicts side effect probabilities, and trains the network.
- Input: Drug embeddings for a combination and ground-truth clinical side effect labels from HODDI.
- Processing: Attention Pooling merges the drugs into a single 128-dimensional vector, and a 2-layer Classification MLP outputs probabilities across 4,569 side effect types.
- Training: Optimized using Focal Loss to focus on rare, dangerous side effects, trained with the Adam optimizer under 5-Fold Cross-Validation.



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